

Integration

The Indefinite Integral

The opposite of a derivative is the **anti-derivative** (indefinite integral). The indefinite integral of a function $f(x)$ is denoted $\int f(x)dx$

Constant of Integration

The quantity (C) in the following equation is called **Constant of Integration**.

$$\int f'(x)dx = f(x) + C$$

This constant is of great importance in scientific applications.

Some Basic Integration Formulas

$$1) \int x^m dx = \frac{x^{m+1}}{m+1} + c \quad m \neq -1$$

$$2) \int \frac{dx}{x} = \ln(x) + c$$

$$3) \int e^x dx = e^x + c$$

Integration of trigonometric functions

$$1) \int \sin(x) dx = -\cos(x) + c$$

$$2) \int \cos(x) dx = \sin(x) + c$$

$$3) \int \tan(x) dx = \ln|\sec(x)| + c$$

$$4) \int \cot(x) dx = \ln|\sin(x)| + c$$

Integration of hyperbolic functions

$$1) \int \sinh(x) dx = \cosh(x) + c$$

$$2) \int \cosh(x) dx = \sinh(x) + c$$

$$3) \int \tanh(x) dx = \ln|\cosh(x)| + c$$

$$4) \int \coth(x) dx = \ln|\sinh(x)| + c$$

Properties of the indefinite integral

- 1- A constant factor can be taken outside the integral sign:

$$\int af(x)dx = a \int f(x)dx \quad , \quad a \equiv \text{constant}$$

- 2- Integral of the sum or difference of functions (additivity)

$$\int [f(x) \pm g(x)]dx = \int f(x)dx \pm \int g(x)dx$$

Example

$$1) \int x^2 dx = \frac{x^3}{3} + c$$

$$2) \int (x^3 + x) dx = \frac{x^4}{4} + \frac{x^2}{2} + c$$

$$3) \int \frac{dx}{\sqrt[3]{x^2}} = \int x^{\frac{-2}{3}} dx = \frac{x^{\frac{1}{3}}}{\frac{1}{3}} + c = 3x^{\frac{1}{3}} + c$$

$$4) \int \frac{2}{x^2} dx = \int 2x^{-2} dx = 2 \frac{x^{-1}}{-1} + c = -\frac{2}{x} + c$$

$$5) \int 5 dx = 5x + c$$

Exercise (9-1)

Evaluate the following integrals

$$(1) \int \sqrt{x} dx$$

$$(4) \int \sqrt{5} x^{-\frac{2}{3}} dx$$

$$(2) \int (4 + \frac{3}{7}x - 6x^2) dx$$

$$(5) \int -\frac{2}{x^3} dx$$

$$(3) \int (1 - t)^2 dt$$

$$(6) \int \left(\frac{3x^2 - 5x}{x} \right) dx$$

Method of Integration

In some infinite integration problems, we find integrals that are **not in basic formulas**, but it is sometimes possible to **convert such integrals to basic formulas** by using some methods.

(1) Integration by Substitution

It is also called the “**Change of variable**” method of integration, and it is one of the important methods for finding the integration of a function and is used directly when we **fail to find the integration** by using the Basic Integration Formulas.

1- The integrals in the form $\int (ax + b)^m dx$

Put $u = ax + b$
 $du = adx$

$$\begin{aligned}\therefore \int (ax + b)^m dx &= \int u^m \left(\frac{1}{a} du\right) = \frac{1}{a} \int u^m du \\ &= \frac{1}{a} \frac{u^{m+1}}{(m+1)} + c = \frac{(ax+b)^{m+1}}{a(m+1)} + c\end{aligned}$$

Example

(1) $\int (2x - 5)^7 dx$

Put $u = 2x - 5 \rightarrow du = 2dx \rightarrow dx = \frac{du}{2}$

$$\begin{aligned}\therefore \int (2x - 5)^7 dx &= \int (u)^7 \frac{du}{2} = \frac{1}{2} \int u^7 du \\ &= \frac{1}{2} \frac{u^8}{8} + c = \frac{(2x-5)^8}{16} + c\end{aligned}$$

(2) $\int (3x - 7)^{\frac{1}{2}} dx$

Put $u = 3x - 7 \rightarrow du = 3dx \rightarrow dx = \frac{du}{3}$

$$\begin{aligned}\therefore \int (3x - 7)^{\frac{1}{2}} dx &= \int (u)^{\frac{1}{2}} \frac{du}{3} = \frac{1}{3} \int u^{\frac{1}{2}} du \\ &= \frac{1}{3} \frac{u^{\frac{3}{2}}}{\frac{3}{2}} + c = \frac{(3x-7)^{\frac{3}{2}}}{\frac{9}{2}} + c\end{aligned}$$

2- The integrals in the form $\int (f(x))^n f'(x) dx$

Put $u = f(x)$
 $du = f'(x) dx$

$$\begin{aligned}\therefore \int [f(x)]^n f'(x) dx &= \int u^n du \\ &= \frac{u^{n+1}}{n+1} + c = \frac{[f(x)]^{n+1}}{n+1} + c\end{aligned}$$

Example

(1) $\int 6x^2(2x^3 - 6)^4 dx$

Put $u = 2x^3 - 6 \rightarrow du = 6x^2 dx \rightarrow dx = \frac{du}{6x^2}$

$$\begin{aligned}\therefore \int 6x^2(2x^3 - 6)^4 dx &= \int 6x^2 u^4 \frac{du}{6x^2} \\ &= \int u^4 du = \frac{u^5}{5} + c = \frac{(2x^3-6)^5}{5} + c\end{aligned}$$

$$(2) \int x^3(x^4 - 2)^5 dx$$

Put $u = x^4 - 2 \rightarrow du = 4x^3 dx \rightarrow dx = \frac{du}{4x^3}$

$$\begin{aligned} \therefore \int x^3(x^4 - 2)^5 dx &= \int x^3 u^5 \frac{du}{4x^3} \\ &= \frac{1}{4} \int u^5 du = \frac{1}{4} \frac{u^6}{6} + c = \frac{(x^4 - 2)^6}{24} + c \end{aligned}$$

$$(3) \int \cos(3x + 7) dx$$

Put $u = 3x + 7 \rightarrow du = 3dx \rightarrow dx = \frac{du}{3}$

$$\begin{aligned} \therefore \int \cos(3x + 7) dx &= \int \cos(u) \frac{du}{3} \\ &= \frac{1}{3} \int \cos(u) du = \frac{1}{3} \sin(u) + c = \frac{1}{3} \sin(3x + 7) + c \end{aligned}$$

The integrals in the form $\int \frac{f'(x)}{f(x)} dx$

Put $u = f(x)$
 $du = f'(x) dx$

$$\begin{aligned} \therefore \int \frac{f'(x)}{f(x)} dx &= \int \frac{du}{u} \\ &= \ln(u) + c = \ln(f(x)) + c \end{aligned}$$

Example

$$(1) \int \frac{3}{5x} dx$$

Put $u = 5x \rightarrow du = 5dx \rightarrow dx = \frac{du}{5}$

$$\therefore \int \frac{3}{5x} dx = \int \frac{3}{u} \frac{du}{5} = \frac{3}{5} \int \frac{du}{u} = \frac{3}{5} \ln(u) + c = \frac{3}{5} \ln(5x) + c$$

$$(2) \int \frac{x}{2+3x^2} dx$$

Put $u = 2 + 3x^2 \rightarrow du = 6x dx \rightarrow dx = \frac{du}{6x}$

$$\therefore \int \frac{x}{2+3x^2} dx = \int \frac{x}{u} \frac{du}{6x} = \frac{1}{6} \int \frac{du}{u} = \frac{1}{6} \ln(u) + c = \frac{1}{6} \ln(2 + 3x^2) + c$$

$$(3) \int \frac{4}{(5x-3)} dx$$

Put $u = 5x - 3 \rightarrow du = 5dx \rightarrow dx = \frac{du}{5}$

$$\therefore \int \frac{4}{(5x-3)} dx = \int \frac{4}{u} \frac{du}{5} = \frac{4}{5} \int \frac{du}{u} = \frac{4}{5} \ln(u) + c = \frac{4}{5} \ln(5x - 3) + c$$

4- The integrals in the form $\int e^{f(x)} f'(x) dx$

Put $u = f(x)$
 $du = f'(x) dx$

$$\begin{aligned}\therefore \int e^{f(x)} f'(x) dx &= \int e^u du \\ &= e^u + c = e^{f(x)} + c\end{aligned}$$

Example

(1) $\int (x-1)e^{x^2-2x+1} dx$

Put $u = x^2 - 2x + 1 \rightarrow du = (2x - 2)dx \rightarrow dx = \frac{du}{2(x-1)}$

$$\begin{aligned}\therefore \int (x-1)e^{x^2-2x+1} dx &= \int (x-1)e^u \left(\frac{du}{2(x-1)} \right) \\ &= \frac{1}{2} \int e^u du = \frac{1}{2} e^u + c = \frac{1}{2} e^{x^2-2x+1} + c\end{aligned}$$

(2) $\int (\cos(x) - 1)e^{\sin(x)-x} dx$

Put $u = \sin(x) - x \rightarrow du = (\cos(x) - 1)dx \rightarrow dx = \frac{du}{(\cos(x)-1)}$

$$\begin{aligned}\therefore \int (\cos(x) - 1)e^{\sin(x)-x} dx &= \int (\cos(x) - 1)e^{\sin(x)-x} \left(\frac{du}{(\cos(x)-1)} \right) \\ &= \int e^u du = e^u + c = e^{\sin(x)-x} + c\end{aligned}$$

5- The integrals in the form $\int e^{(ax+b)} dx$

Put $u = ax + b$
 $du = adx$

$$\begin{aligned}\therefore \int e^{(ax+b)} dx &= \int e^u \left(\frac{1}{a} du \right) = \frac{1}{a} \int e^u du \\ &= \frac{1}{a} e^u + c = \frac{1}{a} e^{(ax+b)} + c\end{aligned}$$

Example

(1) $\int 2e^{6x-1} dx$

Put $u = 6x - 1 \rightarrow du = 6dx \rightarrow dx = \frac{du}{6}$

$$\begin{aligned}\therefore \int 2e^{6x-1} dx &= 2 \int e^u \left(\frac{du}{6} \right) \\ &= \frac{2}{6} \int e^u du = \frac{1}{3} e^u + c = \frac{1}{3} e^{6x-1} + c\end{aligned}$$

Exercise (9-2)**Evaluate the following integrals**

(1) $\int \frac{x^2}{\sqrt{x^3+1}} dx$

(5) $\int \frac{\cos(x)}{2 \sin(x)+3} dx$

(2) $\int x^3 e^{4x^4} dx$

(6) $\int x \sqrt{x^2+1} dx$

(3) $\int \frac{(1+3x)}{\sqrt{2x+3x^2}} dx$

(7) $\int (3x - x^3)^5 (1 - x^2) dx$

(4) $\int (1 + \sin(t))^2 \cos(t) dt$

(8) $\int \sin(x) e^{1+\cos(x)} dx$

(2) Integration by Parts

It is one of the important ways to perform integrations and depends on the following result: If f and g are differentiable functions, then

$$D[f(x)g(x)] = f(x).g'(x) + g(x).f'(x)$$

This means that $g(x)$ is a differential inverse of the right-hand path, then

$$f(x).g(x) = \int f(x).g'(x)dx + \int g(x).f'(x)$$

Rearrange, this becomes.

$$\int f(x)g'(x)dx = f(x).g(x) - \int g(x).f'(x)dx$$

Put

$$u = f(x) \quad \rightarrow \quad dv = g'(x)dx$$

$$du = f'(x)dx \quad \rightarrow \quad v = g(x)$$

The formula can be written by finding

$$\int u dv = uv - \int v du$$

Example

1- $\int x e^x dx$

Put $u = x \quad \Rightarrow \quad dv = e^x dx$

$$du = dx \quad \Rightarrow \quad v = e^x$$

$$\int x e^x dx = \int u dv = uv - \int v du$$

$$\begin{aligned} \int x e^x dx &= x e^x - \int e^x dx \\ &= x e^x - e^x + c = e^x(x - 1) + c \end{aligned}$$

2- $\int \ln(x) dx$

Put $u = \ln(x) \quad \Rightarrow \quad dv = dx$

$$du = \frac{1}{x} dx \quad \Rightarrow \quad v = x$$

$$\int \ln(x) dx = \int u dv = uv - \int v du$$

$$= x \ln(x) - \int x \cdot \frac{1}{x} dx$$

$$= x \ln(x) - x + c = x(\ln(x) - 1) + c$$

$$3- \int e^x \sin(x) dx$$

$$\text{Put } u = \sin(x) \Rightarrow dv = e^x dx$$

$$du = \cos(x) dx \Rightarrow v = e^x$$

$$\begin{aligned} \int e^x \sin(x) dx &= \int u dv = uv - \int v du \\ &= e^x \sin(x) - \int e^x \cos(x) dx \end{aligned}$$

Repeat the same process by taking

$$u = \cos(x) \Rightarrow dv = e^x dx$$

$$du = -\sin(x) dx \Rightarrow v = e^x$$

$$\begin{aligned} \int e^x \sin(x) dx &= e^x \sin(x) - [e^x \cos(x) + \int e^x \sin(x) dx] \\ &= e^x \sin(x) - e^x \cos(x) - \int e^x \sin(x) dx \end{aligned}$$

Therefore

$$2 \int e^x \sin(x) dx = e^x (\sin(x) - \cos(x))$$

Thus, the general formula for integration is:

$$\int e^x \sin(x) dx = \frac{1}{2} e^x (\sin(x) - \cos(x)) + c$$

$$4- \int \sin^{-1}(x) dx$$

$$\text{Put } u = \sin^{-1}(x) \Rightarrow dv = dx$$

$$du = \frac{1}{\sqrt{1-x^2}} dx \Rightarrow v = x$$

$$\begin{aligned} \int \sin^{-1}(x) dx &= (\sin^{-1}(x))(x) - \int x d(\sin^{-1}(x)) \\ &= x \sin^{-1}(x) - \int x \frac{1}{\sqrt{1-x^2}} dx \\ &= x \sin^{-1}(x) + \sqrt{1-x^2} + c \end{aligned}$$

Exercise (9-3)

Evaluate the following integrals

$$(1) \int x e^{-3x} dx$$

$$(3) \int x^2 e^x dx$$

$$(2) \int x \sin(x) dx$$

The Definite Integral

The area bounded by the x axis, the vertical lines $x = a$ and $x = b$ and the function $f(x)$ is denoted with a **definite integral**, $\int_a^b f(x)dx$

The area is signed, that is, if $f(x)$ is negative, then the area is negative.

Properties of the definite integral

If $f(x)$ and $g(x)$ are two continuous functions on the integration interval $a \leq x \leq b$, then

$$(1) \int_a^a f(x)dx = 0$$

$$(2) \int_a^b f(x)dx = -\int_b^a f(x)dx$$

$$(3) \int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx, \quad a \leq c \leq b$$

Example

$$(1) \int_1^3 x^2 dx = \left[\frac{x^3}{3} \right]_1^3 = \frac{(3)^3}{3} - \frac{1^3}{3} = \frac{26}{3}$$

$$(2) \int_0^1 x dx = \left[\frac{x^2}{2} \right]_0^1 = \frac{1^2}{2} - 0 = \frac{1}{2}$$

$$(3) \int_0^{\frac{\pi}{2}} \cos(x) dx = [\sin(x)]_0^{\frac{\pi}{2}} = \sin\left(\frac{\pi}{2}\right) - \sin(0) = 1$$

Exercise (9-4)

Evaluate the following integrals

$$(1) \int_1^2 \int 3x \, dx$$

$$(3) \int_0^3 (x^3 - 4x + 1) dx$$

$$(2) \int_{-2}^3 \int (4 - x^2) \, dx$$